Yash Rampuria

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EDUCATION

Stanford University [2025 - 2027]
Master of Science in Aeronautics and Astronautics (**Robotics** Focus) GPA: **4.0**

Indian Institute of Technology Bombay [2021 - 2025]
Bachelor of Technology (Honors) in Mechanical Engineering | **Ranked 4th** / 190+ students GPA: **9.63 / 10**

PROFESSIONAL EXPERIENCE

Skydio | Autonomy Engineer - Computer Vision [Jun '26 – Sep '26]
Worked on low-latency pose estimation models and **bridging the sim2real gap** for fixed wing autonomous drones
- Built a purely sim-trained feature point extractor achieving sub-pixel keypoint accuracy and < 5cm real-world pose error
- Utilized feature-level domain adaptation, CycleGANs and conditional diffusion models to improve OOD generalization

Honda R&D | Computer Vision Researcher (Self-Driving) [Jun '25 – Aug '25]
Internship on developing LiDAR based SLAM algorithms for autonomous cars | Received full-time offer
- Developed a point-to-point IMU-aided ICP **fusion** pipeline for terrain height mapping in feature-poor environments
- Reduced ICP misalignment by over **80%** by tuning frequency, leveraging IMU priors, ground removal and clustering

Amazon.com, Inc. | Software Development Engineer [May '24 – Jul '24]
Improved & automated testing for mid-mile transportation accounting | Received full-time offer (certificate)
- Implemented a **regression** testing framework for transport accounting using **Java JSON** configurations and TestNG
- Deployed 8+ approved code reviews to reduce user latency in integration testing by automating test data generation
- Utilized AWS - SQS/SNS and **DynamoDb** and presented an SOP to senior engineers for 16+ dataflows in pipelines

ROBOTICS AND COMPUTER VISION

CTO - Autonomous Systems & Perception | IITB Racing Driverless [Mar '22 – Jul '25]
Led the L4 autonomous racing division in a team of **100+** students representing India at Formula Student AI, UK
- Stood **4th** overall, **3rd** in Business Plan and became the only Asian team to run 10 laps at Formula Student AI '25
- Built a novel **LiDAR - Mono/Stereo Camera - IMU** sensor fusion system, robust to edge cases and sensor failure
- Achieved 22% higher accuracy by developing a novel depth estimation deep neural network for LiDAR based vision
- Reduced object detection and classification latency from **100ms to 7ms** with a C++ **TensorRT** engine for **YOLOv11**

Research Assistant at Autonomous Systems Lab, Stanford | | Guide: Prof. Marco Pavone [Jul '26 – Sep '26]
Developed GGraph: GPU-Accelerated Graph Library for Robot Deployment and Learning | Submission to ICRA 2027
- Built a **GPU-resident navigation-graph** library for tensorized world representations, updating a 1M-node graph in 62ms
- Achieved **27x** throughput over CPU graph builders by batching 1024 robots, cutting RL training from **24h to 2h**
- Trained PPO graph-attention exploration policies that beat baselines by 6% in path length at 97% sucessful coverage
- Deployed the sim-trained policy **zero-shot** on a Boston Dynamics Spot and added 2ms vectorized loop-closure correction

AWARDS AND PUBLICATIONS

- Awarded the highest institute-wide honour of '**Person of the Year** - Technical' among 14,000+ students
- Yash Rampuria, et al., "IIT Bombay Racing Driverless: **Autonomous Driving** Stack for Formula Student AI," (link)
- Yash Rampuria, et al., "Electron beam melting of Inconel alloys for aerospace applications: A review of microstructure and mechanical properties," COPEN-13th Int'l Conference on Precision, Micro, Meso, Nano Eng. (link)

SKILLS AND COURSES

- **Robotics**: Foundation Models / VLMs, Navigation, State estimation, Optimization, Planning, Reinforcement Learning
- **Software**: Python, C++, Java, ROS, AWS, Testing, Data Structures and Algorithms, GPU, PyTorch, OpenCV, MATLAB
- **Soft Skills**: Leadership, Problem Solving, Initiative, Ownership, Communication, Creativity, Debugging, Collaboration